MERCK MANUAL
Professional Version

Deep Venous Thrombosis (DVT)

Full Review: Jan 2026 By ***James D. Douketis****, MD, McMaster University | Peer reviewed by* ***Jonathan G. Howlett****, MD, Cumming School of Medicine, University of Calgary*
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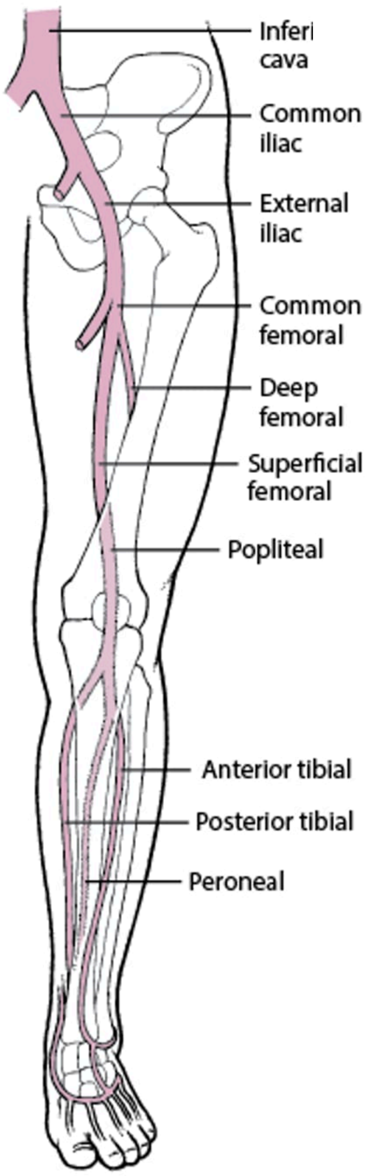
Deep venous thrombosis (DVT) is clotting of blood in a deep vein of an extremity (usually calf or thigh) or the pelvis. DVT is the primary cause of pulmonary embolism. DVT results from conditions that impair venous return, lead to endothelial injury or dysfunction, or cause hypercoagulability. DVT may be asymptomatic or cause pain and swelling in an extremity; pulmonary embolism is an immediate complication. Diagnosis is by history, physical examination, and testing, typically with duplex ultrasound. D-Dimer testing is sometimes used when DVT is suspected; a negative result helps to exclude DVT, whereas a positive result is nonspecific and requires additional testing to confirm DVT. Treatment is with anticoagulants. Prognosis is generally good with prompt, adequate treatment. Common long-term complications include venous insufficiency with or without the post-thrombotic syndrome.

DVT occurs most commonly in the lower extremities or pelvis (see figure Deep Veins of the Legs). DVT is less common in deep veins of the upper extremities (< 5% of DVT cases) (1).

Deep Veins of the Legs



Lower extremity DVT is much more likely to cause pulmonary embolism (PE) than upper extremity DVT, possibly because of the higher clot burden (2). Most proximal DVTs involve the femoral or popliteal veins and up to 38% extend more proximally to involve the iliofemoral veins (2, 3, 4). DVT of the distal or calf veins usually involves the posterior tibial or peroneal veins. Distal or calf vein DVT is less likely to be a source of large emboli but can propagate to the proximal thigh veins and from there cause PE. Approximately 50% of patients with DVT have occult PE, and at least 30% of patients with PE have demonstrable DVT (5).

Pearls & Pitfalls

- Approximately 50% of patients with DVT have occult pulmonary emboli.

General references

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Etiology of Deep Venous Thrombosis

Many factors can contribute to DVT (see table Risk Factors for Deep Venous Thrombosis and Pulmonary Embolism). Cancer is a risk factor for DVT, particularly in older patients and in patients with recurrent thrombosis. Among all patients with cancer, approximately 1 to 5% develop a venous thromboembolism (VTE) annually (1, 2, 3). The association is strongest for pancreatic, brain, hematologic, lung, and ovarian cancers (2, 4). Occult cancers may be present in patients with apparently idiopathic DVT, but extensive evaluation patients for tumors is not recommended unless patients have risk factors for specific cancers or symptoms suggestive of an occult cancer (5).

Etiology references

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Pathophysiology of Deep Venous Thrombosis

Classically, thrombosis occurs due to a combination of impaired blood flow (stasis), endothelial injury, and hypercoagulability (Virchow triad).

Deep venous thrombosis (DVT) usually begins in venous valve cusps. Thrombi consist of thrombin, fibrin, and red blood cells with relatively few platelets (red thrombi); without treatment, thrombi may propagate proximally to the lungs.

Lower extremity DVT most often results from:

- Impaired venous return (eg, in immobilized patients)
- Endothelial injury or dysfunction (eg, after leg fractures)
- Hypercoagulability

Upper extremity DVT most often results from (1):

- Endothelial injury due to central venous catheters, pacemakers, or injection drug use

Upper extremity DVT occasionally occurs as part of superior vena cava (SVC) syndrome (compression or invasion of the superior vena cava by a tumor and causing symptoms such as facial swelling, dilated neck veins, and facial flushing) or results from a hypercoagulable state or subclavian vein compression at the thoracic outlet (1). The compression may be due to a normal or an accessory first rib or fibrous band (thoracic outlet syndrome) or occur during strenuous arm activity (effort thrombosis, or Paget-Schroetter syndrome, which is rare).

Complications

Common complications of DVT include:

- Post-thrombotic syndrome (chronic venous insufficiency)
- Pulmonary embolism

Much less commonly, acute massive DVT leads to phlegmasia alba dolens or phlegmasia cerulea dolens, both of which, unless promptly diagnosed and treated, can result in venous gangrene.

In **phlegmasia alba dolens,** a rare complication of DVT during pregnancy, the leg turns milky white or paler than the surrounding skin. Pathophysiology is unclear, but edema may increase soft-tissue pressure beyond capillary perfusion pressures, resulting in tissue ischemia and venous gangrene. Phlegmasia alba dolens may progress to phlegmasia cerulea dolens.

In **phlegmasia cerulea dolens,** massive iliofemoral venous thrombosis causes near-total venous occlusion; the leg becomes ischemic, extremely painful, and cyanotic or violaceous. Pathophysiology may involve complete stasis of venous and arterial blood flow in the lower extremity because venous return is occluded or massive edema cuts off arterial blood flow. Venous gangrene may result.

Infection rarely develops in venous clots. Jugular vein suppurative thrombophlebitis (Lemierre syndrome), a bacterial (usually anaerobic) infection of the internal jugular vein and surrounding soft tissues, may follow tonsillopharyngitis and is often complicated by bacteremia and sepsis. In septic pelvic thrombophlebitis, pelvic thromboses develop postpartum and become infected, causing intermittent fever. Suppurative (septic) thrombophlebitis, a bacterial infection of a superficial peripheral vein, comprises infection and clotting and usually is caused by venous catheterization.

Pathophysiology reference

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Symptoms and Signs of Deep Venous Thrombosis

DVT may occur in ambulatory patients or as a complication of surgery or major medical illness. Among patients who are hospitalized and are at high risk, deep vein thrombosis is often asymptomatic since typical features of swelling, redness, and tenderness may be less prominent in recumbent than ambulatory patients.

When present, symptoms and signs of DVT (eg, vague aching pain, tenderness along the distribution of the veins, edema, erythema) are nonspecific, vary in frequency and severity, and are similar in arms and legs. Dilated collateral superficial veins may become visible or palpable. Calf discomfort elicited by ankle dorsiflexion with the knee extended (Homans sign) occasionally occurs with distal leg DVT but is neither sensitive nor specific. Tenderness, swelling of the whole leg, > 3 cm difference in circumference between calves, pitting edema, and collateral superficial veins may be most specific; DVT is likely with a combination of ≥ 3 in the absence of another likely diagnosis (see table Wells Score: Probability of Deep Venous Thrombosis Based on Clinical Factors).

Low-grade fever may be present; DVT may be the cause of fever without an obvious source, especially in postoperative patients. Symptoms of pulmonary embolism, if it occurs, may include shortness of breath and pleuritic chest pain.

TABLE

Wells Score: Probability of Deep Venous Thrombosis Based on Clinical Factors

Factors
Tenderness along distribution of the veins in calf or thigh
Swelling of entire leg
Calf swelling (> 3 cm difference in circumference between calves, measured 10 cm below the tibial tuberosity)
Pitting edema greater in affected leg
Dilated collateral superficial veins
Active cancer (treatment ongoing or given within the previous 6 months)

Immobilization of lower extremity (eg, due to paralysis, paresis, casting, or recent long-distance travel)
Surgery leading to immobility for > 3 days within the past 4 weeks
Previous DVT
Probability
Probability equals the number of factors, subtracting 2 if another diagnosis is as likely as or more likely than deep venous thrombosis. • High probability: ≥ 3 points • Moderate probability: 1–2 points • Low probability: ≤ 0 points

Based on Wells PS, Owen C, Doucette S, Fergusson D, Tran H. Does this patient have deep vein thrombosis? *JAMA.* 2006;295(2):199-207. doi: 10.1001/jama.295.2.199 and Scarvelis D, Wells PS. Diagnosis and treatment of deep-vein thrombosis. *CMAJ.* 2006;175(9):1087-1092. doi: 10.1503/cmaj.060366

Common causes of asymmetric leg swelling that mimic DVT are:

- Soft-tissue trauma
- Cellulitis
- Compression of a pelvic vein
- Obstruction of a lymphatic vessel in the pelvis
- Popliteal cyst (including Baker cyst) that obstructs venous return

Less common causes include abdominal or pelvic tumors that obstruct venous or lymphatic return.

Symmetric bilateral leg swelling is the typical result of use of medications that cause dependent edema (eg, dihydropyridine calcium channel blockers, estrogen, high-dose opioids), venous hypertension (usually due to right heart failure), and hypoalbuminemia; however, such swelling may be asymmetric if venous insufficiency coexists and is worse in one leg.

Common causes of calf pain that mimic acute DVT include:

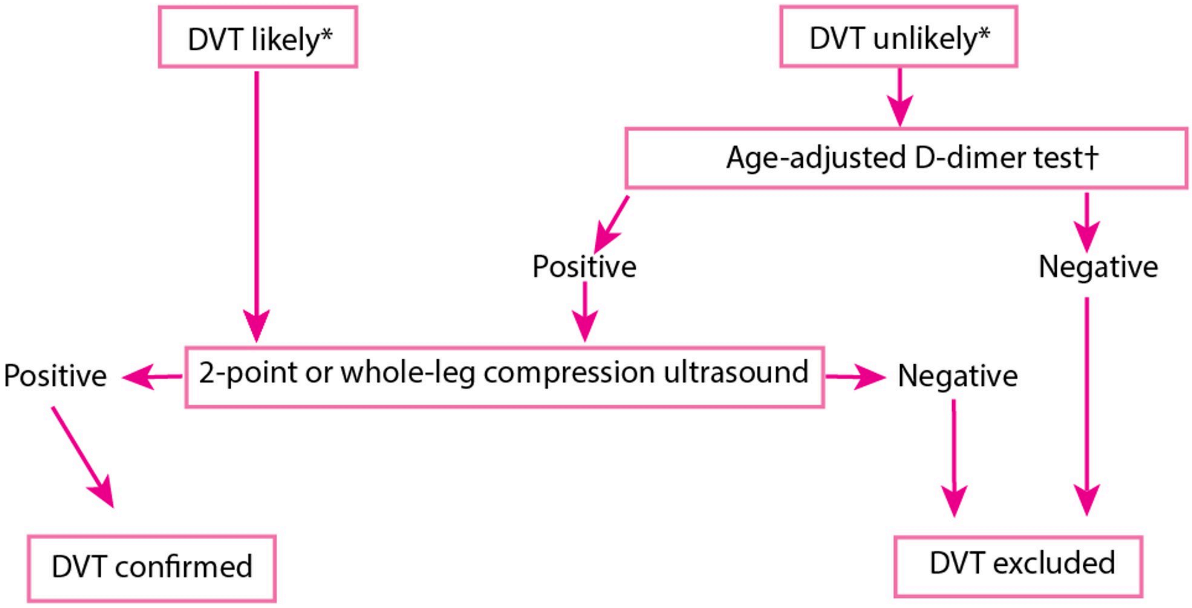
- Venous insufficiency and post-thrombotic syndrome
- Cellulitis that causes painful erythema of the calf
- Ruptured popliteal cyst (pseudo-DVT), which causes calf swelling, pain, and sometimes bruising in the region of the medial malleolus
- Partial or complete tears of the calf muscles or tendons

Diagnosis of Deep Venous Thrombosis

- Ultrasound
- Sometimes D-dimer testing

History and physical examination help determine probability of DVT before testing (see table Wells Score: Probability of Deep Venous Thrombosis Based on Clinical Factors). Diagnosis is typically by ultrasound with Doppler flow studies (duplex ultrasound). The need for additional tests (eg, D-dimer testing) and their choice and sequence depend on pretest probability and sometimes ultrasound results. (See figure One Approach to Testing for Suspected Deep Venous Thrombosis (DVT).)

One Approach to Testing for Suspected Deep Venous Thrombosis (DVT)



* Based on a clinical decision tool such as the Wells criteria.

† The age-adjusted D-dimer threshold is calculated as the patient's age multiplied by 10 ng/mL (10 mcg/L) and is used for patients > 50 years with suspected venous thromboembolism.

Data from Chopard R, Albertsen IE, Piazza G. Diagnosis and Treatment of Lower Extremity Venous Thromboembolism: A Review. *JAMA*. 2020;324(17):1765-1776. doi:10.1001/jama.2020.17272

No single testing protocol is best; one approach is for clinicians to first estimate the probability of a DVT using a validated clinical prediction rule such as the Wells score (1). In patients with a low probability, a negative D-dimer is considered sufficient to exclude DVT and avoid further imaging (2, 3). In patients with a high probability estimated by the clinical prediction rule, or those with a positive D-dimer, compression ultrasound is used to diagnose (or exclude) DVT (4).

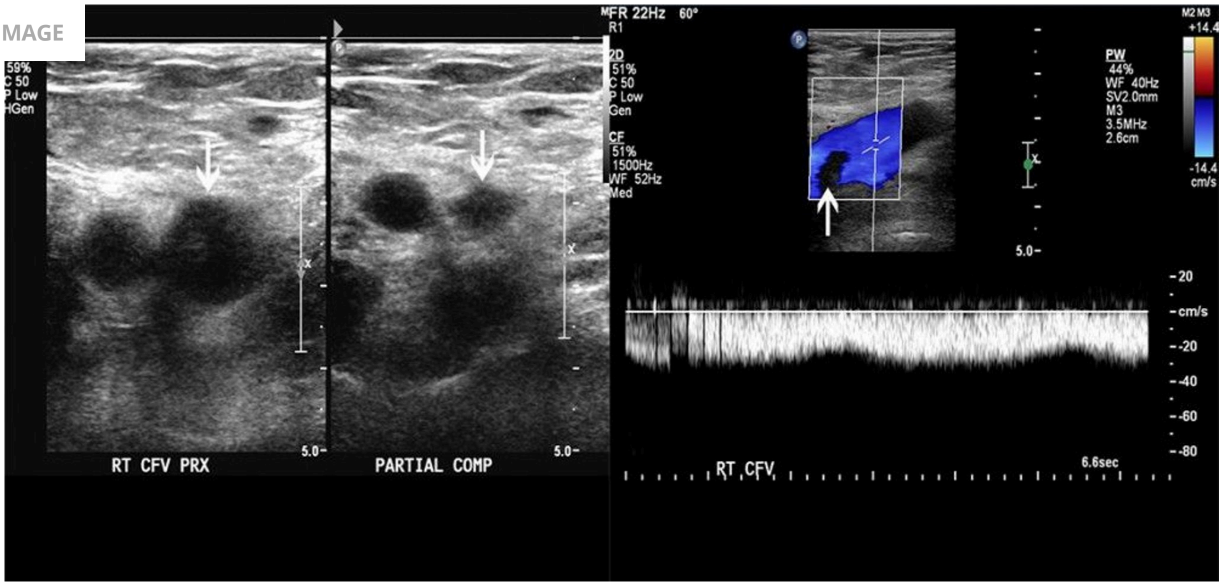
CLINICAL CALCULATORS

DVT Probability: Wells Score System

Ultrasound

Doppler Ultrasound of a Patient with a Thrombus in Femoral Vein

IMAGE



The image on the left shows partial compressibility of the common femoral vein (arrows). The image on the right shows a filling defect on color Doppler flow. These findings are consistent with thrombus.

Ultrasound identifies thrombi by directly visualizing the venous lining and by demonstrating abnormal vein compressibility or, with Doppler flow studies, impaired venous flow. The test is approximately 90% sensitive and 98% specific for lower extremity DVT (5).

D-Dimer

D-Dimer is a byproduct of fibrinolysis; elevated levels suggest recent presence and lysis of thrombi. In contrast to compression ultrasound, D-dimer is highly sensitive (sensitivity approximately 96%) but nonspecific (specificity approximately 37%) (4). A positive test result is nonspecific because levels can be elevated by other conditions (eg, liver disease, trauma, pregnancy, positive rheumatoid factor, inflammation, recent surgery, cancer), and further testing is necessary. D-dimer levels also increase with age, which further decreases the specificity in older patients (6).

Alternative imaging

Contrast-enhanced computed tomographic venography and magnetic resonance venography are other imaging modalities, generally reserved for cases in which ultrasound results are negative or indeterminate and the clinical suspicion for DVT remains high. These imaging tests are less well validated for DVT, are more costly, and may be associated with other complications (eg, related to exposure to radiation and contrast agents).

Testing for pulmonary embolism

If symptoms and signs suggest PE, additional imaging (eg, CT pulmonary angiography or, less often, ventilation/perfusion [V/Q] scanning) is required. For more detail, see Pulmonary Embolism.

Determination of cause

Patients with confirmed DVT and an obvious risk factor (eg, immobilization, surgical procedure, leg trauma), as well as most patients with a first unprovoked DVT need no further testing.

Patients in whom testing to detect thrombophilia may be useful to guide duration of treatment include (7):

- Patients with recurrent unprovoked thrombosis
- Patients with unprovoked thrombosis at an unusual site (e.g., splanchnic, cerebral sinus)
- Women who develop thrombosis within 6 months of starting hormonal therapy (eg, oral contraceptives)

Some evidence suggests that testing for the presence of hypercoagulability in patients with or without clinical risk factors does not predict DVT recurrence (8, 9, 10).

Screening patients with DVT for cancer has a low yield (11). Selective testing guided by complete history and physical examination and basic "routine" tests (complete blood count, chest radiograph, urinalysis, liver enzymes, serum electrolytes, blood urea nitrogen [BUN], creatinine) aimed at detecting cancer is probably adequate. In addition, patients should have any appropriate cancer screening (eg, mammography, colonoscopy) that is due.

Diagnosis references

Treatment of Deep Venous Thrombosis

- Anticoagulation
- Sometimes thrombolytic therapy, surgery, or inferior vena cava filter

It is preferable and safer to prevent DVT than to treat it, particularly in patients who are at high risk. For a complete discussion, see DVT Prevention.

Treatment is aimed primarily at pulmonary embolism prevention and secondarily at symptom relief and prevention of DVT recurrence, chronic venous insufficiency, and post-thrombotic syndrome. Treatment of lower and upper extremity DVT is generally the same.

General supportive measures include pain control with analgesics, which may include short (3- to 5-day) courses of a nonsteroidal anti-inflammatory drug (NSAID). Extended treatment with NSAIDs and aspirin should be avoided because their antiplatelet effects may increase the risk of bleeding complications. In addition, elevation of legs (supported by a pillow or other soft surface to avoid venous compression) is recommended during periods of inactivity (1). Patients may be as physically active as they can tolerate; there is no evidence that early activity increases risk of clot dislodgement and PE and may help to reduce the risk of the post-thrombotic syndrome (2).

Anticoagulation strategies for DVT

(For details on specific medications and their complications, see Medications for Deep Venous Thrombosis.)

There are several strategies for anticoagulation of patients with DVT:

- Initial treatment with an injectable heparin (unfractionated or low molecular weight) followed after several days by long-term treatment with an oral medication (warfarin, a factor Xa inhibitor, or a direct thrombin inhibitor)
- Initial and long-term treatment with a low molecular weight heparin (LMWH)
- Initial and long-term treatment with certain oral factor Xa inhibitors (rivaroxaban or apixaban)

Almost all patients with DVT are treated with anticoagulants (3, 4). The exception is patients with isolated distal thrombi without severe symptoms, in whom serial imaging and anticoagulation are needed only if the thrombus extends. Various anticoagulants are suitable for initial therapy, and the choice of agent is influenced by patient comorbidities (eg, renal dysfunction, cancer), preferences, cost, and convenience.

Inadequate anticoagulation in the first 24 to 48 hours may increase risk of recurrence or of PE. Acute DVT can be treated on an outpatient basis unless severe symptoms require parenteral analgesics, other disorders preclude safe outpatient discharge, or other factors (eg, functional, socioeconomic) might prevent the patient from adhering to prescribed treatments. Dosing regimens for treatment of DVT differ from those used for prevention of thromboembolism in patients with atrial fibrillation.

Appropriate treatment with anticoagulation, particularly warfarin with close international normalized ratio (INR) monitoring, reduces but does not eliminate the risk of post-thrombotic syndrome (5).

Direct Oral Anticoagulants

Direct oral anticoagulants (DOAC), including apixaban, dabigatran, edoxaban, and rivaroxaban, are preferred over warfarin as first-line treatment for DVT and PE, including cancer-associated thromboses. (see table Oral Anticoagulants) (3, 4, 6). Compared to warfarin, these medications have been shown to give similar protection against recurrent DVT and have similar (or with apixaban, perhaps lower) risk of serious bleeding (3). Also, they are given as a fixed dose and thus, unlike warfarin, do not require ongoing laboratory testing. The main disadvantage is the higher cost compared to warfarin and the high cost of DOAC reversal agents in the case of bleeding or need for an urgent surgery or procedure.

For patients who are to start edoxaban (an oral factor Xa inhibitor) or dabigatran (an oral direct thrombin inhibitor), the oral agent is started after 5 days of injectable heparin (7). Alternatively, selected direct oral anticoagulants (rivaroxaban or apixaban) may be initiated immediately without first giving an injectable heparin.

Warfarin

Some patients are initially given an injectable heparin (typically low molecular weight) for 5 to 7 days, followed by longer term treatment with warfarin, which is started within 24 hours after the start of the injectable heparin. Although heparin acts rapidly and provides immediate anticoagulation, warfarin takes about 5 days to achieve a therapeutic effect; hence, heparin is required for 5 to 7 days. Warfarin is the recommended approach for patients with antiphospholipid syndrome (4).

Low molecular weight heparin

For selected patients (eg, with extensive iliofemoral DVT or cancer), continued treatment with a low-molecular-weight heparin rather than switching to an oral agent may be preferred (6).

Fondaparinux, a parenteral factor Xa inhibitor, is sometimes substituted for low molecular weight heparin and can also be used to treat acute DVT.

Duration of Treatment

The initial duration of anticoagulation is 3 to 6 months (3, 4). Patients with transient risk factors for DVT (eg, immobilization, surgery) can usually discontinue anticoagulation after 3 to 6 months. Patients with idiopathic (or unprovoked) DVT with no known risk factors, recurrent DVT, or DVT due to a chronic and persistent risk factor (including cancer and hypercoagulable states such as antiphospholipid syndrome or protein C, protein S, or antithrombin deficiency) should continue anticoagulation indefinitely (extended anticoagulation) for secondary prevention of recurrent DVT and prevention of PE, with periodic re-evaluation of risks and benefits. In some patients, lifelong anticoagulation is needed.

For selected patients who require extended anticoagulation, reduced-dose apixaban or rivaroxaban regimens may be considered. The use of reduced-dose apixaban for patients with cancer-associated DVT and reduced-dose rivaroxaban for patients at high risk for DVT recurrence appear to be as effective as full-dose regimens for extended anticoagulation, while also conferring a lower risk of significant bleeding (1, 8, 9).

Inferior vena cava (IVC) filter

An IVC filter may help prevent PE in patients with lower extremity DVT who have contraindications to anticoagulant therapy (3, 4). An IVC filter is placed in the inferior vena cava just below the renal veins via catheterization of an internal jugular or femoral vein. Some IVC filters are retrievable and can be used temporarily (eg, until contraindications to anticoagulation resolve).

Inferior Vena Cava (IVC) Filter

IMAGE

This CT scan shows an implanted IVC filter.

ZEPHYR/SCIENCE PHOTO LIBRARY

Although IVC filters reduce the risk of acute pulmonary embolism after DVT, they do not reduce mortality and do increase the risk of subsequent DVT, as well as other longer-term complications (eg, venous collaterals can develop, providing a pathway for emboli to circumvent the filter) (4, 10). There is also an increased risk of recurrent DVT. Also, IVC filters can dislodge or become obstructed by a clot. A clotted filter may cause bilateral lower extremity venous congestion (including acute phlegmasia cerulea dolens), lower body ischemia, and acute kidney injury. Treatment for a dislodged filter is removal, using angiographic or, if necessary, surgical methods.

Thrombolytic (fibrinolytic) therapy

Thrombolytic medications, which include alteplase, tenecteplase, and streptokinase, lyse clots and may be more effective than anticoagulation alone in selected patients, but the risk of bleeding is higher than with heparin. Consequently, thrombolytics should be considered only in highly selected patients with DVT, such as those < 60 years with extensive iliofemoral DVT who have evolving or existing limb ischemia (eg, phlegmasia cerulea dolens) and do not have risk factors for bleeding (3).

Catheter-directed thrombolysis has largely replaced systemic administration when used for DVT, and may be considered in some patients to help prevent post-thrombotic syndrome (5).

Surgery

Surgery is rarely needed. However, thrombectomy, fasciotomy, or both are mandatory for phlegmasia alba dolens or phlegmasia cerulea dolens unresponsive to thrombolytics to try to prevent limb-threatening gangrene.

Treatment references

Prognosis for Deep Venous Thrombosis

Even with treatment, lower extremity DVT can have approximately a < 1 to 2% risk of fatal PE (1, 2); death due to upper extremity DVT is very rare. Risk of recurrent DVT is lowest for patients with transient risk factors (eg, surgery, trauma, temporary immobility) and greatest for patients with persistent risk factors (eg, cancer), idiopathic DVT, or incomplete resolution of past DVT (residual thrombus). A normal D-dimer level obtained after anticoagulation is stopped for 3 to 4 weeks may help predict a relatively low risk of DVT or PE recurrence, more so in women than in men (3). Risk of venous insufficiency is difficult to predict. Risk factors for post-thrombotic syndrome include proximal (iliofemoral) thrombosis, recurrent ipsilateral DVT, increased body mass index, older age, and suboptimal anticoagulation (4).

Prognosis references

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Key Points

- **Symptoms and signs are nonspecific, so clinicians must be alert, particularly in high-risk patients.**
- **Low-risk patients may have D-dimer testing, as a normal result essentially excludes deep venous thrombosis (DVT); others should have ultrasound.**
- **Treatment initially is with an injectable heparin (unfractionated or low molecular weight heparin [LMWH]) followed by an oral anticoagulant (warfarin, dabigatran, or a factor Xa inhibitor) or a LMWH; alternatively, the oral factor Xa inhibitors rivaroxaban and apixaban may be used for initial and ongoing treatment.**
- **Duration of treatment is typically 3 to 6 months, depending on the presence and nature of risk factors; certain patients require extended treatment.**

Drug Information for the Topic

MERCK MANUAL
Professional Version

Deep Venous Thrombosis (DVT) Prevention

*Full Review: Jan 2026 By **James D. Douketis**, MD, McMaster University | Peer reviewed by **Jonathan G. Howlett**, MD, Cumming School of Medicine, University of Calgary*
Last updated: May 2026

It is preferable and safer to prevent deep venous thrombosis (DVT) than to treat it, particularly in patients who are at risk. The general principles of DVT prevention are assessing the risk of DVT, assessing the risk and consequences of bleeding, and choosing a DVT prevention approach (1, 2, 3). Preventive measures include:

- Ambulation and prevention of immobility
- Prophylactic anticoagulation (eg, low molecular weight heparin [LMWH], fondaparinux, adjusted-dose warfarin, direct oral anticoagulant)
- Intermittent pneumatic compression

Specific risk assessment tools and recommendations vary among orthopedic surgery patients, nonorthopedic surgery patients, and medical patients.

(See also Deep Venous Thrombosis.)

General references

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Risk Assessment in DVT Prevention

An individualized risk assessment is recommended for hospitalized patients using validated tools to guide prophylaxis decisions.

For **surgical patients**, risk of DVT is based on the specific procedure as well as patient-specific factors. Risk assessment tools, including the Caprini and Rogers scores, are useful for assessing risk after nonorthopedic surgical procedures (see Risk of Deep Venous Thrombosis and Pulmonary Embolism in Surgical Patients) (1). These models incorporate clinical risk factors (eg, reduced mobility, previous DVT, acute infection) to categorize the patient's risk for DVT (eg, low, high).

For **medical patients**, risk of DVT is based on the cause and severity of the underlying illness as well as patient-specific risk factors. Risk assessment tools include the Padua and IMPROVE venous thromboembolism (VTE) scores (2). Bleeding risk can be estimated using the IMPROVE bleeding risk score. Patients being treated in a critical care unit, or those with heart failure, cancer, respiratory failure, prolonged (≥ 3 days) immobilization, hormonal medications, known thrombophilia, prior VTE, age ≥ 70 years) are at higher risk for VTE (3).

TABLE

Risk of Deep Venous Thrombosis and Pulmonary Embolism in Surgical Patients

Risk Category	Approximate Risk of DVT/PE (%)	Caprini Score	Rogers Score	Examples	Preventive Measures
Very Low	< 0.5	0	< 7	Minor procedures (< 30–45 minutes) in patients < 40 years with no risk factors	No mechanical or pharmacologic prophylaxis
Low	1.5	1–2	7–10	Major surgery (>45 minutes) in patients < 40 years Patients 40–74 years with no other risk factors Pregnant or postpartum	IPC

Risk Category	Approximate Risk of DVT/PE (%)	Caprini Score	Rogers Score	Examples	Preventive Measures
				patients	
Moderate	3	3–4	> 10	Patients ≥ 75 years with no other risk factors Family history of VTE Known thrombophilia Major surgery in patients < 40 years with no other clinical risk factors	LMWH, LDUH, or IPC
High	6	≥ 5	—	Major orthopedic surgery (arthroplasty, hip or pelvis fracture)> Acute spinal cord injury Multiple risk factors	LMWH or LDUH, and IPC Extended LMWH after abdominal/pelvic surgery for cancer IPC if high bleeding risk; aspirin + fondaparinux or IPC if heparin

Risk Category	Approximate Risk of DVT/PE (%)	Caprini Score	Rogers Score	Examples	Preventive Measures
					contraindicated

DVT = deep venous thrombosis; IPC = intermittent pneumatic compression; LDUH = low-dose unfractionated heparin; LMWH = low molecular weight heparin; PE = pulmonary embolism.

Data from Gould MK, Garcia DA, Wren SM, et al. Prevention of VTE in nonorthopedic surgical patients: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest.* 2012;141(2 Suppl):e227S-e277S. doi:10.1378/chest.11-2297Ada

Risk assessment references

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Method of DVT Prevention

DVT prophylaxis can involve one or more of the following:

- Mechanical therapy (eg, compression devices or stockings, venous filters)
- Pharmacologic therapy (including low-dose unfractionated heparin, low molecular weight heparins, warfarin, fondaparinux, direct oral anticoagulants)

Choice depends on patient’s risk level for DVT and risk of bleeding, type of surgery (if applicable), projected duration of preventive treatment, contraindications, adverse effects, relative cost, ease of use, and local practice.

In general, patients undergoing major surgery should receive mechanical or pharmacologic prophylaxis, with mechanical prophylaxis preferable in patients with higher risk of bleeding (1, 2). Although major surgeries are not strictly defined, they typically include high-risk procedures such as those involving open abdominal or pelvic surgery, cancer surgery, major trauma, and surgeries with prolonged immobility, whereas minor surgeries include short laparoscopic operations or minor soft tissue procedures. For certain higher-risk procedures both mechanical and pharmacologic prophylaxis are suggested. For patients undergoing major orthopedic surgery, such as total hip or knee arthroplasty or surgery for hip fracture, pharmacologic prophylaxis with direct oral anticoagulants or LMWH, usually for 10 to 35 days, is recommended (1, 3). If bleeding risk is high, mechanical prophylaxis is an option.

For patients undergoing transurethral resection of the prostate, laparoscopic cholecystectomy, and most neurosurgical procedures, pharmacologic prophylaxis is generally not recommended.

Patients acutely or critically ill and hospitalized for nonsurgical conditions are usually given pharmacologic prophylaxis with LMWH or low-dose unfractionated heparin (UFH or LDUH) (4, 5). Mechanical prophylaxis is used if bleeding risk or other factors preclude heparin (4, 5). Anticoagulation is typically discontinued at hospital discharge.

Mechanical therapy for DVT prophylaxis

After surgery, early mobilization is likely to reduce the risk of postoperative DVT.

The benefit of graded compression stockings is questionable except for low-risk surgical patients and selected hospitalized medical patients. However, combining stockings with other preventive measures may be more protective than any single approach.

Intermittent pneumatic compression (IPC) uses a pump to cyclically inflate and deflate hollow plastic leggings, providing external compression to the lower legs and sometimes thighs. IPC may be used instead of or in combination with anticoagulants after surgery. IPC is recommended for patients undergoing surgery associated with a high risk of bleeding in whom anticoagulant use may be contraindicated (1, 2). IPC is probably more effective for preventing calf DVT than proximal DVT. IPC is contraindicated in some patients who have obesity and may be unable to apply the devices properly.

For patients who are at very high risk of DVT and bleeding (eg, after major trauma), IPC is recommended until the bleeding risk subsides and anticoagulants can be given (1, 2).

Inferior vena cava filters are not indicated for primary prevention of DVT.

Pharmacologic therapy for DVT prophylaxis

Pharmacologic thromboprophylaxis involves use of anticoagulants.

Low molecular weight heparins (LMWHs) are more effective than low-dose UFH for preventing DVT and PE (6, 7, 8). LMWH is the preferred agent for patients undergoing moderate to high-risk nonorthopedic surgeries and for medical patients who are acutely and critically ill (1, 2, 4, 5). For patients undergoing major orthopedic surgery, LMWH may also be used for thromboprophylaxis, although direct oral anticoagulants are generally preferred (3). Enoxaparin 30 mg subcutaneously every

12 hours, dalteparin 5000 units subcutaneously once a day, and tinzaparin 4500 units subcutaneously once a day appear to be equally effective. Fondaparinux 2.5 mg subcutaneously once a day is at least as effective as LMWH in patients who are undergoing nonorthopedic surgery and is possibly more effective than LMWHs after orthopedic surgery (9). Prophylaxis is usually discontinued upon discharge, except after total knee or hip arthroplasty or hip fracture surgery, where it is continued for 14 to 35 days.

Low-dose unfractionated heparin (UFH, or LDUH) can be used for patients undergoing moderate or high-risk nonorthopedic procedures or major orthopedic procedures and for medical inpatients; however, in most situations LMWH is preferred (1, 2, 5). Dosing is 5000 units subcutaneously given 2 hours before surgery and every 8 to 12 hours thereafter for 7 to 10 days or until patients are fully ambulatory. Patients who are bedbound but are not undergoing surgery are given 5000 units subcutaneously every 12 hours until risk factors are reversed.

Direct oral anticoagulants (DOACs, eg, dabigatran, rivaroxaban, apixaban) are preferred over LMWH and UFH for preventing DVT and PE after hip or knee replacement surgery.(1, 3, 7).

Fondaparinux is an alternative to heparins but is not a first-line anticoagulant due to bleeding risk and renal toxicity (2, 3, 5).

Warfarin, using a target international normalized ratio (INR) of 2.0 to 3.0 may be used in orthopedic surgery, but other agents are preferred (1, 3).

The role of aspirin for DVT prophylaxis is largely limited to patients undergoing total hip or knee replacement surgery (1, 3, 10).

With DVT prophylaxis, there is always a risk of bleeding during use of anticoagulants.

DVT prophylaxis in selected populations

For hip and other lower extremity orthopedic surgery, thromboprophylaxis with selected direct oral anticoagulants (eg, rivaroxaban, apixaban), LMWH, fondaparinux, or adjusted-dose warfarin is recommended (1, 3). For patients undergoing total knee replacement and some other patients at high risk in whom anticoagulants cannot be given because of a high risk of bleeding, IPC is also beneficial. For orthopedic surgery, preventive treatment may be started before or after surgery and continued for 14 days after total knee replacement and 28 days after total hip replacement. For hip fracture repair surgery, preventive treatment should be given for 14 to 35 days, depending on the patient's mobility status. Fondaparinux 2.5 mg subcutaneously once a day appears to be more effective to prevent DVT than LMWH for patients undergoing orthopedic surgery but may be associated with an increased risk of bleeding (9).

For elective neurosurgery, mechanical prophylaxis is preferred (1). For neurosurgery or trauma in patients with low to moderate bleeding risk where pharmacologic prophylaxis is used, low-dose LMWH is the first option, or low-dose UFH is recommended.

Preventive treatment is also indicated for patients who have a major medical illnesses that require bed rest (eg, myocardial infarction, ischemic stroke, heart failure). Low-dose UFH or LMWH is effective in patients who are not already receiving IV heparin or thrombolytics; IPC, elastic stockings, or both may be used when anticoagulants are contraindicated. For select patients with cancer who are at high risk (eg, those with advanced pancreatic cancer) who are receiving chemotherapy, primary prophylaxis with LMWH or certain direct oral anticoagulants (apixaban or rivaroxaban) may be considered (11, 12, 13, 14).

Patients who are traveling for long periods (eg, airplane flight > 4 hours), ambulation and calf exercises are recommended over compression stockings, aspirin, or LMWH (5). However, for travelers with 2 DVT risk factors (recent surgery, history of DVT, pregnant or postpartum persons, active malignancy, hormone replacement therapy, obesity), compression stockings or LMWH can be considered.

Treatment references

Key Points

- **Treatment to prevent deep venous thrombosis is required for patients who are bedbound with major illness and/or those undergoing certain surgical procedures.**
- **Early mobilization, leg elevation, and an anticoagulant are the recommended preventive measures; patients who should not receive anticoagulants may benefit from intermittent pneumatic compression devices or elastic stockings.**

Drug Information for the Topic

MERCK MANUAL
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Medications for Deep Venous Thrombosis

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Almost all patients with deep venous thrombosis (DVT) are given anticoagulants and in < 5% of cases thrombolytics. A number of anticoagulants are effective for management of deep venous thrombosis (see also Treatment of Deep Venous Thrombosis).

Anticoagulants

The anticoagulants (see figure Anticoagulants and Their Sites of Action and table Oral Anticoagulants) include:

- Low molecular weight heparins (LMWHs)
- Unfractionated heparin (UFH)
- Factor Xa inhibitors: oral (eg, rivaroxaban, apixaban, edoxaban) and parenteral (fondaparinux)
- Direct thrombin inhibitors: oral (dabigatran etexilate) and parenteral (argatroban, bivalirudin, desirudin)
- Warfarin

Oral factor Xa inhibitors and direct thrombin inhibitors are sometimes referred to as direct oral anticoagulants (DOACs). However, there are also parenteral agents that inhibit both factor Xa and thrombin (unfractionated heparin), that inhibit mainly factor Xa (LMWH), or that inhibit only factor Xa (fondaparinux). These agents can be used both for patients with DVT and those with pulmonary embolism (PE).

See Treatment of Deep Venous Thrombosis for a discussion of anticoagulation strategies including selection, initiation, and duration of therapy.

Anticoagulants and Their Sites of Action

LMWH = low molecular weight heparin; TF = tissue factor; UFH = unfractionated heparin.

TABLE

Direct Oral Anticoagulants for Treatment of Venous Thromboembolism

Medication	Dose	Comments
Factor Xa inhibitors		
Apixaban	10 mg twice a day for 7 days Then 5 mg twice a day	Apixaban should not be used for DVT or PE if creatinine clearance is < 30 mL/minute (0.5mL/s/m^2). Dosage may be lowered to 2.5 mg twice a day after 6–12 months of treatment.
Edoxaban	60 mg once a day If creatinine clearance is 15–50 mL/minute (0.25 to 0.83 mL/s/m^2) or if body	Initial treatment with a low molecular weight heparin is needed for 5 days.

Medication	Dose	Comments
	weight is ≤ 60 kg, 30 mg once a day	Edoxaban should not be used if creatinine clearance is < 15 mL/minute (0.25 mL/s/m2).
Rivaroxaban	15 mg twice a day for 21 days taken with food Then 20 mg once a day taken with food	Rivaroxaban should not be used for DVT or PE if creatinine clearance is < 30 mL/minute (0.5mL/s/m^2). Dosage may be lowered to 10 mg once a day after 6–12 months of treatment.
Factor IIa (thrombin) inhibitor		
Dabigatran	150 mg twice a day	Initial treatment with low molecular weight heparin is needed for 5 days. Dabigatran should not be used for DVT if creatinine clearance is < 30 mL/minute (0.5mL/s/m^2).

DVT = deep venous thrombosis; PE = pulmonary embolism.

Factor Xa inhibitors

Rivaroxaban and apixaban can be started as monotherapy immediately upon diagnosis or used in transition from an injectable heparin at any time without overlap (1). Dosing of rivaroxaban is 15 mg orally twice a day for 3 weeks followed by 20 mg orally once a day for a total of 3 to 6 months. Apixaban dosing is 10 mg orally twice a day for 7 days followed by 5 mg orally twice a day for 3 to 6 months.

Before edoxaban is started, an initial 5 to 7 days of treatment with LMWH or UFH is required. Then edoxaban is given 60 mg orally once a day. To transition from an injectable anticoagulant, the factor Xa inhibitor is typically started within 6 to 12 hours after the last dose of a twice-daily LMWH regimen and within 12 to 24 hours after a once-daily LMWH regimen.

There is evidence that apixaban, edoxaban, and rivaroxaban can be used in select patients with cancer-associated venous thromboembolism (VTE) as an alternative to monotherapy with LMWH (2, 3, 4, 5).

Fondaparinux, a parenteral selective factor Xa inhibitor, may be used as an alternative to UFH or LMWH for the initial treatment of DVT or PE. It is given in a fixed dose of 7.5 mg subcutaneously once a day (10 mg for patients > 100 kg, 5 mg for patients < 50 kg). It has the advantage of fixed dosing and is less likely to cause thrombocytopenia.

Direct thrombin inhibitors

Dabigatran 150 mg orally twice a day is given only after an initial 5 days of treatment with LMWH (1). It is typically started within 6 to 12 hours after the last dose of a twice-daily LMWH regimen and within 12 to 24 hours after a once-daily regimen.

Parenteral direct thrombin inhibitors (argatroban, bivalirudin, desirudin) are available but do not have a role in treatment or prevention of DVT or PE. Argatroban may be useful to treat DVT in patients with heparin-induced thrombocytopenia (6).

Vitamin K antagonists (warfarin)

Warfarin is a treatment option for patients with DVT who are not pregnant. It is also an option for patients with severe renal dysfunction. Warfarin is also a second-line option for patients with cancer-associated VTE.

Warfarin can be started, initially at a dose of 5 to 10 mg once daily, immediately along with heparin because it takes about 5 days to achieve desired therapeutic effect. Older patients and patients with a liver disorder typically require lower warfarin doses. Therapeutic goal is an international normalized ratio (INR) of 2.0 to 3.0 (1). INR is monitored weekly for the first 1 to 2 months of warfarin treatment and monthly thereafter; the dose is adjusted to maintain the INR within this range. Patients taking warfarin should be informed of possible medication interactions, including interactions with foods and nonprescription medicinal herbs.

Rarely, warfarin causes skin necrosis in patients with inherited protein C deficiency or protein S deficiency.

Low molecular weight heparins (LMWHs)

A low molecular weight heparin (LMWH, eg, enoxaparin, dalteparin, tinzaparin) can be given on an outpatient basis. Like UFH, LMWHs catalyze the action of antithrombin (which inhibits coagulation factor proteases), leading to inactivation of coagulation factor Xa and, to a lesser degree, factor IIa. LMWHs also have some antithrombin–mediated anti-inflammatory properties, which facilitate clot organization and resolution of symptoms and inflammation. LMWHs are as effective as unfractionated heparin (UFH) for reducing DVT recurrence, thrombus extension, and risk of death due to PE (7). The primary indications for LMWH as a first-line therapy are for DVT in pregnant patients and for breakthrough DVT in patients taking warfarin, in which case it is preferred over DOACs (1, 8, 9).

LMWHs are typically given subcutaneously in a standard weight-based dose (eg, enoxaparin 1.5 mg/kg subcutaneously once a day or 1 mg/kg subcutaneously every 12 hours or dalteparin 200 units/kg subcutaneously once a day). Patients with renal insufficiency may be treated with UFH or with reduced doses of LMWH. Monitoring is not reliable because LMWHs do not significantly prolong the results of global tests of coagulation. Furthermore, they have a predictable dose response, and there is no clear relationship between the anticoagulant effect of LMWH and bleeding. If warfarin is used after initial anticoagulation, treatment with LMWH is continued until full anticoagulation is achieved with warfarin (typically about 5 days). Transition to the oral agents rivaroxaban or apixaban can be done at any time with no overlap. Transition to edoxaban or dabigatran requires at least 5 days of LMWH treatment, but no overlap is needed.

Complications of LMWHs include bleeding, thrombocytopenia, urticaria, and, rarely, thrombosis and anaphylaxis. Inpatients and possibly outpatients should be screened for bleeding with serial complete blood counts and, where appropriate, testing for occult blood in stool.

Unfractionated heparin (UFH)

Unfractionated heparin may be used instead of LMWH for patients who are hospitalized and for patients who have chronic kidney disease with a creatinine clearance 10 to 30 mL/minute, (0.17 to 0.5mL/s/m^2) because UFH is not cleared by the kidneys. UFH is given as a bolus and infusion to achieve full anticoagulation (eg, activated partial thromboplastin time [aPTT] 1.5 to 2.5 times that of the reference range). For outpatients, UFH 333 units/kg initial bolus, then 250 units/kg subcutaneously every 12 hours can be substituted for IV UFH to facilitate mobility; the dose does not need adjustment based on aPTT. If warfarin is used after initial anticoagulation, UFH is continued until full anticoagulation has been achieved with warfarin (usually about 5 days).

Complications of UFH are similar to those of LMWHs, except that heparin-induced thrombocytopenia, which is rare and may be life-threatening, is more common with UFHs. Long-term use of UFH causes hyperkalemia, liver enzyme elevations, and osteopenia. Rarely, UFH given subcutaneously causes skin necrosis. Inpatients and possibly outpatients being given UFH should be screened for bleeding.

Anticoagulant treatment references

Bleeding During Use of Anticoagulants

Bleeding is the most common complication of anticoagulants and ranges on a continuum from minor to severe, life-threatening hemorrhage.

For **minor bleeding** (eg, epistaxis), local measures to stop bleeding (eg, direct pressure) are often sufficient. The anticoagulant is usually not discontinued or reversed unless bleeding becomes more severe.

For **severe bleeding** (eg, heavy gastrointestinal bleeding), the anticoagulant is usually discontinued (at least temporarily) and other measures taken. Bleeding is generally considered severe when it is:

- Heavy (loss of ≥ 2 units of blood in ≤ 7 days)
- In a critical location (eg, intracranial, intraocular)

- In a location where hemostasis is difficult to achieve (eg, small bowel, posterior nasal cavity, lung)

Risk factors for severe bleeding include (1, 2):

- Age ≥ 65 to 75 years
- History of prior gastrointestinal bleeding or stroke
- Recent myocardial infarction
- Coexisting anemia (hematocrit < 30%), end-stage renal disease (serum creatinine > 1.5 mg/dL [115 micromol/L]), or diabetes
- Thrombocytopenia
- Active cancer

Supportive care for severe bleeding includes local measures to stop bleeding (eg, direct pressure, cauterization, injection). Patients with signs and symptoms of volume loss and those with heavy ongoing bleeding may require intravenous fluid resuscitation and packed red blood cell transfusions. These measures are sufficient for many bleeding episodes.

In patients with **life-threatening and/or ongoing bleeding** or **bleeding in a critical location,** clinicians also consider giving:

- Reversal agents
- Clotting factors (eg, prothrombin complex concentrate, fresh frozen plasma)
- Antifibrinolytics

However, by definition these agents are prothrombotic, and the risks of continued bleeding should be balanced with the increased risk of thrombosis.

Anticoagulant reversal

Many of the anticoagulants have specific reversal agents. If these are unavailable or ineffective, clotting factors, typically in the form of 4-factor prothrombin complex concentrate or sometimes fresh frozen plasma, can be given. Some anticoagulants can be removed by hemodialysis or have absorption blocked by activated charcoal.

With the **heparins,** bleeding can be stopped or slowed with protamine (3). It is more effective on unfractionated heparin (UFH) than on low molecular weight heparin (LMWH) because protamine only partially neutralizes LMWH inactivation of factor Xa. During all infusions, patients should be observed for hypotension and a reaction similar to an anaphylactic reaction. Because UFH given IV has a half-life of 30 to 60 minutes, protamine is typically not given to patients who have received UFH > 60 to 120 minutes beforehand) or is given at a reduced dose based on the amount of heparin estimated to be remaining in plasma, based on the half-life of UFH.

Warfarin anticoagulation can be reversed with vitamin K (3). If hemorrhage is severe, prothrombin complex concentrate should be given; fresh frozen plasma may be used if prothrombin complex concentrate is unavailable. Selected patients with overanticoagulation (INR 5 to 9) who are neither actively bleeding nor at increased risk of bleeding can be managed by omitting 1 or 2 warfarin doses and monitoring INR more frequently, then giving warfarin at a lower dose.

With **dabigatran**, a humanized monoclonal antibody idarucizumab is an effective antidote to bleeding (3). If the agent is not available, 4-factor prothrombin complex concentratecan be given. Hemodialysis also may help because dabigatran is not highly protein bound. Oral activated charcoal is an option if the last dose of dabigatran was within 2 hours.

With **factor Xa inhibitors,** andexanet alfa is an antidote available in the United States; however, its use is restricted in part due to its high cost (3, 4). Fondaparinux anticoagulation can theoretically also be reversed with andexanet alfa although this has not been studied in research trials. If andexanet alfa is unavailable, 4-factor prothrombin complex concentrate may be considered. Oral activated charcoal is an option in patients who took an oral factor Xa inhibitor within a few hours of presentation (8 hours for rivaroxaban, 6 hours for apixaban, and 2 hours for edoxaban). Hemodialysis is not effective for removal of the oral factor Xa inhibitors.

Clotting factors

Clotting factors are available in the form of:

- Prothrombin complex concentrate
- Fresh frozen plasma
- Individual clotting factors

Prothrombin complex concentrate is available in several forms. Three-factor prothrombin complex concentrate contains high levels of factors II, IX, and X, and 4-factor prothrombin complex concentrate adds factor VII; both also contain protein C and protein S. Prothrombin complex concentrate can be unactivated or activated, in which some of the factors have been cleaved to their active forms. Four-factor prothrombin complex concentrate is preferred as it tends to be more effective at reversing bleeding than the 3-factor form (3, 5). If 3-factor prothrombin complex concentrate is used, fresh frozen plasma can also be given because fresh frozen plasma contains factor VII. Typical dose is 50 units/kg IV. Because evidence of benefit is uncertain and risk of clotting is significant, prothrombin complex concentrates should be reserved for life-threatening bleeding.

Fresh frozen plasma contains all the clotting factors but only at normal plasma levels. It is typically used only if prothrombin complex concentrate is unavailable; there is no evidence it is effective in bleeding due to factor Xa inhibitors (3).

Individual clotting factors such as activated recombinant factor VII are available but are not thought to be helpful for anticoagulant-related bleeding.

Antifibrinolytics and other agents

Antifibrinolytic agents (tranexamic acid and aminocaproic acid) can also be tried; however, their use has not been studied for reversal of bleeding in patients taking anticoagulants.

Resumption of anticoagulation after bleeding

Clinical judgment is necessary when deciding whether to permanently stop or lower the dose of anticoagulant.

If a patient has almost completed their treatment course of anticoagulant and has a severe bleeding episode, the anticoagulant can be stopped. However, if a patient has just started or is mid-way through their treatment course and has a severe bleed, the decision of whether to stop or reduce the dose of anticoagulant is not as straightforward and should be made in consultation with a multidisciplinary team and keeping in mind the patient's priorities.

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Thrombolytic (Fibrinolytic) Agents

Thrombolytic agents, which include alteplase, tenecteplase, and streptokinase, lyse clots and may be more effective than heparin alone in selected patients with DVT, but the risk of bleeding is higher than with heparin alone. For patients with DVT, a clinical trial showed that catheter-directed thrombolytic therapy did not reduce the incidence of post-thrombotic syndrome compared with conventional anticoagulant therapy (1). However, thrombolytic agents can be considered only in highly selected patients with DVT (2). Patients who may benefit from thrombolytic agents include those with extensive iliofemoral DVT who are younger (< 60 years) and do not have risk factors for bleeding. Thrombolytic therapy should be given stronger consideration in patients with extensive DVT who have evolving or existing limb ischemia (eg, phlegmasia cerulea dolens).

Catheter-directed thrombolysis is preferred over systemic thrombolysis for patients with extensive DVT in whom thrombolysis is used (3).

Bleeding, if it occurs, is most often at the site of arterial or venous puncture sites. This complication can be treated by stopping the thrombolytic agent and doing mechanical compression or surgical repair of the puncture site. Life-threatening bleeding is treated with cryoprecipitate and fresh frozen plasma in addition to stopping the thrombolytic agent.

Thrombolytic treatment references

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Drug Information for the Topic